

RISK BASED TEST PLAN

Digital Wallet Transfer Limits
and Account Tiers

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FINANCIAL CONTROL • CONCURRENCY • TIME BOUNDARIES • EXEMPTION SECURITY

Executive Summary

This plan defines a risk based strategy for validating transfer limits across tiers, transfer types, lifecycle states and time boundaries. It prioritises financial control failures that could permit excess transfers, misstate a customer's remaining allowance or create an unsafe exemption.

Highest risk controls	Primary test technique
Backend limit enforcement and atomic reservation	API, service and controlled concurrency testing
Pending, failed and reversed transfer accounting	State transition and ledger reconciliation
Verified own linked account exemption	Identity-state decision table and negative security tests
Midnight reset, scheduling and tier transitions	Controllable business clock and lifecycle testing

Test strategy at a glance

- Full boundary coverage where behaviour changes, representative sampling where only configuration changes.
- Pairwise coverage for lower risk combinations, dedicated scenarios for concurrency, reversals, scheduling and tier changes.
- Backend records are the source of truth, client validation is treated as advisory.
- Working assumptions enable design to proceed but must not be mistaken for approved product rules.

1. Purpose

The objective is to confirm that users cannot exceed applicable per transaction or daily limits, that reservations and releases remain correct throughout the transfer lifecycle, and that all channels report the same authoritative allowance.

- Validate Tier 1, Tier 2 and Tier 3 rules.
- Prevent direct API, duplicate request and race-condition bypasses.
- Protect the own-account exemption with secure ownership verification.
- Apply consistent rules to scheduled transfers, cross-midnight events and tier changes.
- Verify accurate errors, notifications and audit trails.

2. Scope

In scope	Out of scope unless confirmed
Wallet-to-wallet, external bank and verified own linked account transfers	Incoming transfer limits
Immediate and scheduled transfers	Cash withdrawals and merchant payments
Per transaction and daily enforcement, pending reservations	International transfers and foreign exchange
Failures, cancellations, reversals, midnight reset and concurrency	Undescribed bank-specific or regulatory limits
Tier changes, direct API attempts, UI, notifications and auditability	KYC execution itself, except its tier outcome
Backend enforcement and idempotency	Fraud, sanctions and AML beyond limit interaction

3. Current Limit Rules

Tier	Per transaction limit	Daily limit
Tier 1	20,000	50,000
Tier 2	200,000	500,000
Tier 3	No stated cap	5,000,000

Stated behaviour: Pending transfers count toward the daily limit. Failed or reversed transfers restore the applicable amount. Transfers to verified own linked bank accounts are exempt. Scheduled transfers execute at 8:00 AM. A tier upgrade takes effect immediately.

These statements contain unresolved ambiguities. Section 4 records the product questions and provisional assumptions required to make expected results testable.

4. Open Questions and Working Assumptions

The following assumptions permit test design to proceed. They are provisional and require formal Product approval, especially for P0 controls. The question numbering is retained as a decision log.

4.1 Currency and amount calculation

Q1. In which currency are the limits defined?

Working assumption: The wallet supports one configured local currency and all stated limits use it.

Q2. If multiple currencies are supported, is there one limit per currency or one converted daily allowance?

Working assumption: All transfers are converted to a configured base currency before evaluation.

Q3. Are transfer fees included in the limit calculations?

Working assumption: Only principal consumes the stated limit; fees remain subject to balance validation.

Q4. What is the smallest permitted amount and decimal precision?

Working assumption: The smallest valid amount is one minor currency unit.

Q5. Which rounding rule applies?

Working assumption: The backend uses fixed-precision decimal arithmetic and the currency standard minor unit.

Q6. How are zero, negative, malformed or excessively large amounts handled?

Working assumption: Reject through server-side validation before reservation.

4.2 Daily limit ownership and calculation

Q7. Is the daily limit shared between wallet and bank transfers?

Working assumption: They consume one shared allowance.

Q8. Is the limit per user, wallet, account or currency?

Working assumption: It applies to the sending wallet; multiple-wallet behaviour needs confirmation.

Q9. Are exact limit values inclusive?

Working assumption: Yes; only amounts greater than a limit are rejected.

Q10. Does Tier 3 allow one transfer up to its remaining daily allowance?

Working assumption: Yes; there is no separate transaction cap.

Q11. Do regulatory, fraud, bank or channel limits still apply to Tier 3 and exempt transfers?

Working assumption: Yes; exemption covers only these stated limits.

Q12. Can an administrator override a limit?

Working assumption: No, unless an authorised and audited process is defined.

Q13. Can remaining allowance be negative?

Working assumption: No; reservation is atomic and remaining allowance never falls below zero.

Q14. Are there velocity limits on transfer count?

Working assumption: None within this feature unless Product confirms otherwise.

4.3 Midnight reset and timezone

Q15. Which timezone determines midnight?

Working assumption: A fixed platform business timezone, enforced by the backend.

Q16. Is the backend clock authoritative?

Working assumption: Yes; device time changes cannot trigger reset.

Q17. How are travel, daylight saving and different device timezones handled?

Working assumption: They do not change the fixed business window.

Q18. What happens to a request submitted exactly at midnight?

Working assumption: Server receipt time determines the window.

Q19. What if initiated before midnight but received after midnight?

Working assumption: It belongs to the new day.

Q20. Is reset a job or derived from timestamps?

Working assumption: An authoritative limit service derives usage by business date; missed jobs cannot alter eligibility.

Q21. Should the displayed allowance update automatically after midnight?

Working assumption: Yes, without requiring logout.

4.4 Pending transfers and reservations

Q22. At what stage is the amount reserved?

Working assumption: Atomically when the backend accepts an authorised transfer.

Q23. Do transfers awaiting PIN, OTP or biometric approval consume limit?

Working assumption: No.

Q24. Which statuses count as pending?

Working assumption: Accepted processing states that can still complete.

Q25. If a transfer starts before midnight and remains pending after it, which day counts?

Working assumption: The original reservation business date.

Q26. Does it count against every day it remains pending?

Working assumption: No.

Q27. What if it remains unresolved for days?

Working assumption: Retain original reservation until final status or reconciliation.

Q28. Can users cancel pending transfers and when is allowance restored?

Working assumption: Release once after backend-confirmed cancellation.

Q29. What if a transfer completes after an erroneous release?

Working assumption: Raise reconciliation and operational alert; do not silently permit excess usage.

Q30. Is an idempotency key required?

Working assumption: Yes; retry returns the original result without another transfer or reservation.

4.5 Simultaneous transfers and race conditions

Q31. What if two requests fit individually but exceed the remainder together?

Working assumption: Atomic check and reservation allows one and rejects the other.

Q32. How is processing order decided?

Working assumption: The first committed reservation wins.

Q33. Should both fail if their combined amount is too high?

Working assumption: No, unless batch behaviour is explicitly defined.

Q34. How are requests from multiple devices or app instances handled?

Working assumption: All channels use one central atomic ledger.

Q35. What if immediate and scheduled transfers compete?

Working assumption: The first atomic reservation succeeds; the other is rejected if allowance is insufficient.

Q36. How are repeated taps and client retries handled?

Working assumption: Client suppression plus backend idempotency.

4.6 Failed transfers and reversals

Q37. Which final statuses restore reserved limit?

Working assumption: Every conclusively unsuccessful final state releases once.

Q38. How quickly is allowance restored after failure?

Working assumption: Within the same atomic transition or an idempotent recovery.

Q39. What if processor succeeds but wallet times out?

Working assumption: Keep pending until reconciled; do not release prematurely.

Q40. Are partial reversals supported?

Working assumption: If supported, restore only the reversed portion.

Q41. What if yesterday's transfer is reversed after today's reset?

Working assumption: Adjust the original ledger without lifting today above its normal cap.

Q42. Can reversal produce allowance above the tier daily limit?

Working assumption: No.

Q43. What if the same reversal event arrives twice?

Working assumption: Idempotent processing restores once.

Q44. What if reversal and reset occur simultaneously?

Working assumption: Serialise or protect transactionally.

Q45. What if reversal occurs after a tier change?

Working assumption: Release the original reservation but cap availability under the current tier.

Q46. How are refunds, chargebacks and admin adjustments treated?

Working assumption: No limit effect unless mapped to a prior reservation.

4.7 Own linked bank account exemption**Q47. How is ownership verified?**

Working assumption: A trusted identity service and strong identity link; a user-entered name is insufficient.

Q48. Is account-name matching sufficient?

Working assumption: No.

Q49. How are name variations handled?

Working assumption: Use trusted unique identity rather than visual text comparison.

Q50. Are joint accounts considered own accounts?

Working assumption: No, unless Product and Compliance approve.

Q51. Are business, virtual and foreign-currency accounts exempt?

Working assumption: Only when verified under an approved rule.

Q52. Does exemption begin immediately when an account is added?

Working assumption: Only after verification succeeds.

Q53. What if verification expires or is revoked?

Working assumption: New transfers lose exemption immediately.

Q54. Is eligibility evaluated at creation or processing?

Working assumption: At backend final submission; scheduled transfers revalidate at execution.

Q55. What if linked account is removed while transfer is pending?

Working assumption: Accepted transfer retains classification, pending Product and Compliance confirmation.

Q56. Do balance, fraud, compliance and bank rules still apply?

Working assumption: Yes.

Q57. Do exempt transfers appear in the usage ledger?

Working assumption: Yes, clearly marked as exempt.

Q58. Does a pending exempt transfer reserve allowance?

Working assumption: No, but evidence of exemption is retained.

4.8 Scheduled transfers

Q59. Which timezone determines 8:00 AM?

Working assumption: The fixed platform business timezone.

Q60. Is limit reserved at scheduling or execution?

Working assumption: At execution.

Q61. Are funds reserved at scheduling?

Working assumption: No; validate balance at execution.

Q62. Which tier applies?

Working assumption: The tier effective at execution.

Q63. What if allowance is exhausted at 8:00 AM?

Working assumption: Fail and notify; do not roll to another day.

Q64. Should an unsuccessful scheduled transfer retry later that day?

Working assumption: No, unless retry policy is explicitly defined.

Q65. What if balance is insufficient?

Working assumption: Fail without consuming allowance.

Q66. Can scheduled transfers execute on weekends and holidays?

Working assumption: Submit at 8:00 AM; downstream processing may remain pending.

Q67. What if scheduler is unavailable at 8:00 AM?

Working assumption: Recover once with idempotency and intended order.

Q68. How are multiple due transfers prioritised?

Working assumption: A deterministic order with atomic reservation.

Q69. What if only some fit?

Working assumption: Evaluate individually in order; eligible ones succeed.

Q70. Can users edit, pause or cancel?

Working assumption: Cancel before execution; edits trigger revalidation.

Q71. Is own-account exemption revalidated at execution?

Working assumption: Yes.

Q72. Can a retry execute a scheduled transfer twice?

Working assumption: No; a unique execution key prevents duplication.

4.9 Tier upgrades and changes

Q73. What event makes a KYC upgrade effective?

Working assumption: Persisted successful backend KYC decision.

Q74. What does “immediately” mean?

Working assumption: Available to the next request without logout.

Q75. Does upgrade increase today’s allowance?

Working assumption: Yes; existing usage is deducted from the new tier limit.

Q76. Which tier applies during authorisation?

Working assumption: The tier at atomic limit check.

Q77. Which tier applies to an accepted pending transfer?

Working assumption: It retains the acceptance decision and reservation.

Q78. Are pending transfers reevaluated after upgrade?

Working assumption: No; upgrade applies to new decisions.

Q79. Which tier applies to a scheduled transfer created before upgrade?

Working assumption: The tier at execution.

Q80. What if update and transfer arrive simultaneously?

Working assumption: Transactional ordering decides and is audited.

Q81. Are downgrades possible?

Working assumption: Assume yes if KYC is revoked or corrected; behaviour must be defined.

Q82. What if prior usage exceeds a new lower limit?

Working assumption: Completed transfers stand; block additional limited transfers until reset.

Q83. What happens to pending and scheduled transfers after downgrade?

Working assumption: Pending retains reservation; scheduled uses lower tier at execution.

Q84. Are tier changes audited and communicated?

Working assumption: Yes.

4.10 Error handling and user experience

Q85. Are transaction-limit and daily-limit violations separate error codes?

Working assumption: Yes.

Q86. What must a daily-limit rejection contain?

Working assumption: Stable code, accurate remainder, currency and next reset time.

Q87. Should transaction-limit errors show that limit?

Working assumption: Yes, identify the actual violated rule.

Q88. Does displayed remainder include pending reservations?

Working assumption: Yes.

Q89. What if displayed allowance becomes stale before submission?

Working assumption: Backend recalculates and client refreshes from response.

Q90. Should submit be disabled when a known limit is exceeded?

Working assumption: Client may validate early; backend remains mandatory.

Q91. Are messages localised and accessible?

Working assumption: Yes, including amount formatting and screen-reader announcements.

Q92. Which notifications are required?

Working assumption: Scheduled failure, background rejection, reversal and tier change; active errors appear in session.

4.11 Security, consistency and auditability

Q93. Are limits enforced entirely by the backend?

Working assumption: Yes; client checks are advisory.

Q94. What if the limit service is unavailable?

Working assumption: Fail closed with a retryable service error.

Q95. How is consistency maintained across backend instances?

Working assumption: A central transactional ledger or equivalent atomic datastore.

Q96. What audit records are required?

Working assumption: Transfer ID, user, tier, limit window, prior usage, reserve/release amount, exemption reason, transition and timestamp.

Q97. What monitoring is required for excess usage?

Working assumption: Alert on negative remainder, duplicate release or excess usage.

Q98. Must mobile, web and API show the same allowance?

Working assumption: Yes, from one backend source of truth.

5. Risk Assessment and Test Priorities

Priority	Risk area	Reason
P0	Daily and per transaction enforcement	Defect may permit prohibited transfers and financial or compliance exposure.
P0	Concurrent processing	Requests may race against one allowance.
P0	Pending reservation	Incorrect accounting may permit excess or block valid transfers.
P0	Own linked account verification	This path bypasses all stated limits.
P0	Failure and reversal restoration	Bad release can reduce allowance or inflate it.
P0	Backend enforcement and idempotency	Direct calls and retries must not bypass controls.
P1	Midnight and timezone boundaries	May cause early, late or duplicate reset.
P1	Scheduled execution	Relevant balance, tier, ownership and allowance can change.
P1	Tier changes in flight	Inconsistent snapshot may approve or reject incorrectly.
P1	Numerical boundaries	Comparison and configuration defects often surface here.
P2	Messages, display and notifications	Incorrect communication harms recovery and trust.

6. What I Would Test First and Why

1. Backend limit enforcement

Confirm the server independently enforces all rules against modified clients and direct API calls.

2. Per transaction and daily boundaries

Test immediately below, exactly at and one minor unit above thresholds to expose comparison and configuration errors.

3. Atomic reservation under concurrency

Submit individually valid but collectively invalid requests; ordinary sequential testing cannot expose this race.

4. Pending transfer accounting

Confirm immediate reservation and prevent a second count at completion.

5. Own linked account exemption

Validate ownership security before validating the bypass.

6. Failure and reversal restoration

Release once, for the right amount, without exceeding the daily cap.

7. Midnight reset

Exercise before, at and after the authoritative boundary, including pending and reversal events.

8. Scheduled transfers

Revalidate balance, tier, ownership and allowance at execution.

9. Tier transitions

Change tier during authorisation, checking and pending processing.

10. Experience and audit

Reconcile errors, notifications, display and backend audit records.

7. Automation Approach

Test area	Automation approach
Limit boundaries	Data-driven service/API tests below, equal to and above each threshold
Tier configuration	Parameterised tests driven by tier data
Shared daily usage	Sequences combining wallet and external bank transfers
Pending reservations	State validation across accepted, pending, completed and failed
Concurrency	Synchronised parallel requests against one remainder
Failures and reversals	Full, partial, delayed and duplicate event transitions
Midnight reset	Injected or controllable business clock
Scheduled transfers	Scheduler/service tests with simulated execution time
Tier upgrades	Controlled changes at lifecycle checkpoints
Idempotency	Repeated keys and simulated network retries
Exemption rules	Eligible, unverified, revoked and unavailable ownership states
Regression	Compact P0/P1 suite for each relevant release

8. Manual and Exploratory Testing

- Message clarity, accessibility, localisation and currency formatting.
- Background allowance changes on mobile and across devices showing different local time.
- Slow networks, timeouts, repeated taps and interrupted authentication.
- Notification behaviour for scheduling, reversal and tier changes.
- Reconciliation among transfer history, allowance display, notification and backend record.
- Unexpected status transitions and external processor conditions that are not deterministic.
- User understanding of exemption and scheduled-transfer failure.

9. Coverage Strategy

The Cartesian product is intentionally reduced through equivalence partitioning, boundary analysis, pairwise selection, decision tables, state transitions and dedicated high-risk scenarios. Full repetition is used only when expected behaviour changes.

9.1 Coverage dimensions

Dimension	Representative values
Tier	Tier 1, Tier 2, Tier 3
Transfer type	Wallet, external bank, verified own linked bank, scheduled
Limit state	No usage, partial, exact remainder, one minor unit above, exhausted
Amount state	Below, exact and above transaction limit
Transfer state	Created, awaiting authorisation, pending, completed, failed, reversed
Timing	Normal, before/at/after midnight, 8:00 AM execution
Tier state	Stable, upgraded before/during, downgraded if supported
Request pattern	Single, duplicate, simultaneous
Ownership	Unlinked, linked unverified, verified, revoked

9.2 Amount equivalence classes

Class	Expected result
Zero or negative	Reject as invalid
Below both applicable limits	Accept if other controls pass
Exactly transaction limit	Accept
One minor unit above transaction limit	Reject for transaction limit
Exactly remaining daily allowance	Accept
One minor unit above remainder	Reject for daily limit
Tier 3 within remainder	Accept without separate transaction cap
Tier 3 above remainder	Reject for daily limit
Verified own account	Exempt from stated limits
Unverified linked account	Treat as ordinary external transfer

9.3 Matrix reduction rules

- Full numerical boundaries on Tier 1.
- Targeted configuration checks on Tier 2.
- Tier 3 focus on absence of transaction cap and continued daily enforcement.
- Selected wallet plus external-bank sequences prove a shared bucket.
- A separate ownership decision path validates exemption security.
- Dedicated concurrency, reset, reversal, scheduling and tier-transition scenarios.
- Pairwise coverage for secondary dimensions.
- Repeat across every tier only when the expected result changes.

10. Representative Coverage Matrix

Scenario	Tier 1	Tier 2	Tier 3	Wallet	External	Own linked
Below limits	Full	Sample	Sample	Yes	Yes	Exemption path
Exact transaction cap	Full	Sample	N/A	Yes	Pairwise	N/A
Above transaction cap	Full	Sample	N/A	Yes	Pairwise	N/A
Exact daily remainder	Full	Sample	Full	Yes	Yes	N/A
Above daily remainder	Full	Sample	Full	Yes	Yes	Only if verified
Daily exhausted	Full	Pairwise	Full	Yes	Yes	Allowed if valid
Pending reservation	Full	Pairwise	Pairwise	Yes	Yes	No reservation
Concurrency race	Full	Pairwise	Pairwise	Yes	Yes	Separate interaction
Midnight reset	Full	Pairwise	Pairwise	Yes	Yes	Audit only
Failure/reversal	Full	Pairwise	Pairwise	Yes	Yes	No inflation
Scheduled execution	Full	Pairwise	Pairwise	Yes	Yes	Revalidate
Tier change in flight	T1→T2	T2→T3	N/A	Pairwise	Pairwise	Separate interaction

Full = all relevant boundaries and states. Sample = representative values confirming configuration. Pairwise = interaction coverage without complete permutation.

11. Core Limit Decision Table

Own verified	Within transaction cap	Within daily remainder	Expected result
Yes	Any	Any	Allowed under stated exemption, subject to balance, fraud and compliance
No	Yes	Yes	Allowed
No	No	Yes	Reject for transaction limit
No	Yes	No	Reject for daily limit
No	No	No	Reject using Product-defined precedence
Verification unavailable	Any	Any	Do not grant exemption; fail closed or treat as normal transfer per confirmed rule

Decision required: Product must define error precedence when both the per transaction and daily limit are exceeded.

12. Entry Criteria

- Critical P0 ambiguities are answered or assumptions formally accepted.
- Currency, tier limits, statuses and state transitions are available.
- Test users exist for all tiers, with verified and unverified linked accounts.
- Time, scheduler, failure, pending and reversal behaviours can be controlled or simulated.
- Reliable transaction, limit and audit records are accessible.
- Concurrent requests and idempotency keys are supported by the test interface.

13. Exit Criteria

- All planned P0 and P1 scenarios executed.
- No unresolved critical or high-severity defect in enforcement, reservation, exemption, reset, reversal or concurrency.
- Tier boundaries pass and concurrency cannot exceed allowance.
- Duplicate requests and events do not duplicate financial or limit adjustments.
- Scheduled transfers apply confirmed execution-time rules.
- Backend and supported clients show consistent remaining allowance.
- Decisions are documented and critical automated regression is stable.

14. Key Risks and Mitigations

Risk	Mitigation
Concurrent requests exceed daily limit	Atomic check/reservation verified by synchronised requests
Pending transfer counted twice	State-transition and ledger reconciliation
Failure never restores allowance	Automated failure and recovery validation
Duplicate reversal restores twice	Idempotent event-processing tests
Cross-day reversal inflates allowance	Original-ledger adjustment and cap invariant
Device time triggers reset	Authoritative backend clock and timezone manipulation
Unverified account gets exemption	Ownership-state security validation
Scheduled transfer bypasses current rules	Complete revalidation at execution
Tier change creates inconsistent decision	Controlled change at each processing stage
Direct API bypass	Service-side enforcement tests
Displayed value differs from backend	Cross-channel reconciliation
Scheduler retry duplicates transfer	Unique execution key and retry tests

15. Test Deliverables

- Risk Based Test Plan.
- Top 20 Prioritised Test Cases.
- Automated regression results where applicable.
- Manual and exploratory evidence.
- Defect reports with reproducible evidence.
- Requirements clarification and assumption log.
- Final summary covering coverage, defects, residual risk and release recommendation.

16. Conclusion

This plan provides deep coverage of the financial decision logic without creating hundreds of repetitive cases. Numerical behaviour is covered through boundaries and equivalence classes, configuration is sampled across tiers, lower-risk interactions are handled pairwise, and high-risk lifecycle events receive dedicated scenarios.

The earliest and deepest testing targets concurrency bypass, pending and reversal accounting, exemption security, reset boundaries, scheduled execution and in-flight tier changes. Together, these controls protect financial correctness, regulatory intent and customer trust.